

Reinforcement Learning

Lecture Notes

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These notes accompany the interactive visualization *Repair or Replace*, available at
https://sml-fs26.github.io/classic_rl/repair-or-replace/

The concepts developed below map one-to-one onto its scenes.

1 Motivation

Many decisions a manager faces are sequential. What is decided today changes what must be decided tomorrow, and the consequences unfold over weeks and months rather than at once. The outcomes are uncertain on top: the same decision may turn out well one time and badly the next. Picture the scenario behind these notes. You operate a delivery business with a single van, and each week you make exactly one call about that machine: keep running it and collect the week's profit, at the risk of a breakdown and of gradual wear; repair it, paying for a week in the shop so that it fails less often afterwards; or replace it outright, a large expense that buys a fresh start. Every option trades a short-term cost against a long-term payoff, and chance interferes at every step.

Reinforcement learning is a framework for computing decision strategies that maximize the reward accumulated over time in precisely such problems. One running example, the repair-or-replace problem of the visualization, carries everything.

2 Markov decision processes

The first step is a precise model of the decision problem. The standard model is the *Markov decision process*.¹

Definition 2.1 (Markov decision process). A *Markov decision process* (MDP) is a quadruple, a package of four ingredients, (S, A, T, γ) , where S is the set of *states* the system can be in; A is the set of *actions* available to the decision maker; T is the *transition dynamics*, which describes how the system reacts when an action is taken; and $\gamma \in [0, 1]$, the *discount factor*, is a number between 0 and 1. (γ is the Greek letter gamma.)

The sets S and A are defined a priori: they are fixed in advance, part of the problem description, and writing them down is the modeler's first task. The transition dynamics is then defined on top of them. It is a *randomized function* T that consumes a state s and an action a and produces a new state s' and a reward r :

$$(s', r) = T(s, a).$$

¹The name records that the next state depends only on the current state and action, not on the past.

In words: feed the current state and the chosen action into the dynamics, and out come the state the system lands in next and the reward, a number measuring the immediate gain or loss of that step.

Remark 2.2. The transition dynamics is *random*: feeding the same state and action (s, a) in twice may produce different new states and different rewards.²

The discount factor γ is just a number between 0 and 1; its use will be explained later.

Example 2.3 (the repair-or-replace MDP). You manage a one-van delivery fleet; the visualization calls the van *Old Bessie*. Each week the van sits at one of four wear levels, and you make exactly one call:

$$S = \{\text{HEALTHY, WORN, SHAKY, FAILING}\}, \quad A = \{\text{RUN, SERVICE, REPLACE}\}.$$

In words: four wear levels, from factory-fresh to nearly dead, and three possible calls (*SERVICE* is the “repair” of the title). Rewards are weekly net money, in plain units. The numbers governing *RUN* are:

state	weekly profit	breakdown odds
HEALTHY	+95	2%
WORN	+72	8%
SHAKY	+40	28%
FAILING	+16	55%

Under *RUN* the van drives the route and earns the profit in the table, unless it breaks down, with the odds in the table: the week then books a single net reward equal to the profit minus a breakdown cost of 280 (from *HEALTHY*: $95 - 280 = -185$), and the van ends the week at *FAILING*. Even without a breakdown, wear tends to slip by one or two levels over the weeks. Under *SERVICE* the van spends the week in the shop: reward -50 , no deliveries, and the wear often improves; from *WORN*, the van returns to *HEALTHY* 95% of the time. Under *REPLACE* the reward is -130 , and the next week starts with a brand-new van at *HEALTHY*, whatever the old one looked like. The dynamics is random in the sense of Remark 2.2: feeding (*WORN, RUN*) into T may return (*SHAKY, +72*) one week and (*FAILING, -208*) another. There is no final week (the fleet runs indefinitely), and the visualization fixes $\gamma = 0.9$.

3 Policies and trajectories

An MDP describes the world the manager operates in; it does not say how to operate. That is the job of a policy.

Definition 3.1 (policy). A *policy* defines how to act at each state. Formally, a policy is a function $\pi: S \rightarrow A$: it takes a state s and returns the action $\pi(s)$ to be played whenever the system is in s . (π is the Greek letter pi.)

Executing a policy on the MDP, week after week, generates a record of everything that happened. That record is the central object of these notes.

Definition 3.2 (trajectory). The *trajectory* of a policy π is the sequence

$$s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_i, a_i, r_i, \dots$$

²Scene “You run the fleet” of the interactive [visualization](#) lets you sample these dynamics.

produced by executing π on the MDP starting from state s_1 . In words: states, actions, and rewards alternate, and the subscript counts the weeks. The mechanics are as follows. The first action results from applying the policy to the first state, $a_1 = \pi(s_1)$. The first reward and the second state result from applying the transition dynamics to that pair, $(s_2, r_1) = T(s_1, a_1)$. Then $a_2 = \pi(s_2)$, the dynamics produce $(s_3, r_2) = T(s_2, a_2)$, and so on without end.

Remark 3.3. The trajectory is *random*: starting again from the same s_1 and executing the same policy may produce a different trajectory, because the transition dynamics is random (and so, possibly, is the policy, if one allows policies that randomize their choices).

Example 3.4 (a trajectory of the fleet). Consider the playbook π with $\pi(\text{HEALTHY}) = \text{RUN}$, $\pi(\text{WORN}) = \text{SERVICE}$, and $\pi(\text{SHAKY}) = \pi(\text{FAILING}) = \text{REPLACE}$. One execution of π from $s_1 = \text{HEALTHY}$ produced³

$s_1 = \text{HEALTHY},$	$a_1 = \text{RUN},$	$r_1 = +95,$
$s_2 = \text{WORN},$	$a_2 = \text{SERVICE},$	$r_2 = -50,$
$s_3 = \text{HEALTHY},$	$a_3 = \text{RUN},$	$r_3 = +95,$
$s_4 = \text{HEALTHY},$	$a_4 = \text{RUN},$	$r_4 = +95,$

after which $s_5 = \text{WORN}$ and the walk continues, one week per step, forever. In words: the van started healthy and earned 95 but slipped to worn; a 50 service restored it; two profitable runs followed; by week five it was worn again. Remark 3.3 is at work here: another execution of the same π from the same s_1 might open with a first-week breakdown, $s_1 = \text{HEALTHY}$, $a_1 = \text{RUN}$, $r_1 = -185$, $s_2 = \text{FAILING}$.

4 The goal: accumulating reward

A policy is good if it produces trajectories that accumulate as much reward as possible. The first task is to make “accumulate” precise.

Definition 4.1 (cumulative reward). Take one particular trajectory $s_1, a_1, r_1, s_2, a_2, r_2, \dots$. Its *cumulative reward*, also called its *return*, is

$$G_1 = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots + \gamma^{i-1} r_i + \dots \quad (4.1)$$

In words: add up the rewards of all weeks, forever (that is what the dots mean), scaling the reward of week i by γ^{i-1} : week one counts in full, week two is scaled by γ , week three by γ^2 , and so on. (G stands for *gain*.)

Here the discount factor plays its very important role: it specifies how much we care about future rewards. With γ close to 0, the scaling factors die out almost immediately and only near-term rewards matter; with γ close to 1, they fade slowly and rewards far in the future count almost as much as immediate ones. Managers will recognize the device: it is the discounting of future cash flows familiar from net-present-value calculations, and, as with the NPV of a perpetuity, the shrinking factors keep the never-ending total finite. With the visualization’s $\gamma = 0.9$, next week’s reward counts 90% of this week’s, the week after 81%, and so on.⁴ For the trajectory of Example 3.4, formula (4.1) gives $G_1 = 95 + 0.9 \cdot (-50) + 0.9^2 \cdot 95 + 0.9^3 \cdot 95 + \dots = 95 - 45 + 76.95 + 69.255 + \dots$.

³Scene “The trajectory” of the interactive [visualization](#) replays such a tape week by week.

⁴Scene “Return over the van’s life” of the interactive [visualization](#) has a slider for γ .

The same accounting can be started at any week, not only the first. For an arbitrary week i , define G_i as the cumulative reward from week i on:

$$G_i = r_i + \gamma r_{i+1} + \gamma^2 r_{i+2} + \dots \quad (4.2)$$

In words: G_i is what the trajectory accumulates from week i onward, with the discounting restarted there. Comparing (4.2) at weeks i and $i + 1$ yields a recurrence that carries the rest of these notes:

$$G_i = r_i + \gamma G_{i+1}. \quad (4.3)$$

In words: what is accumulated from week i on equals this week's reward plus γ times what is accumulated from week $i + 1$ on.

Remark 4.2. G_i is *random*. Because of the randomness of the transition dynamics, G_i comes out differently after every execution of the policy starting from state s_i .

A quantity that changes from run to run cannot be maximized directly; we average it instead.

Definition 4.3 (expectation). The *expectation* $\mathbb{E}[G_i]$, read “the expected value of G_i ,” is the average of G_i over a sufficiently large set of trajectories: execute the policy many times, always starting from the same first state s_1 (for the fleet: a HEALTHY van), record the value of G_i of each execution, and average the records.

We can now state the ultimate objective of reinforcement learning: *find the policy π that maximizes $\mathbb{E}[G_1]$* , the policy whose trajectories accumulate, on average, the most reward. A policy achieving this maximum is called *optimal*. Note the contrast with G_i itself: $\mathbb{E}[G_i]$ is a single real number that does not depend on the trajectory, because the averaging has removed the randomness. Therefore $\mathbb{E}[G_i]$ is *not* random.

5 The Q -function and the Bellman equation

How can the best policy be discovered? The classical route goes through an auxiliary function that scores every pair of a state and an action.

Definition 5.1 (the Q -function). For a state s and an action a , $Q^*(s, a)$ is the cumulative reward one can expect when executing an optimal policy from state s , where the *first* action taken there is a . In notation,

$$Q^*(s, a) = \mathbb{E}[G_i \mid s_i = s, a_i = a].$$

In words: the expected value of G_i conditioned on s_i being s and a_i being a (the vertical bar is read “given that”); that is, the average of G_i over many executions, counting only those executions that are in state s at week i and take action a there as a one-off, even when an optimal policy would choose differently, acting optimally afterwards. (The star in Q^* marks optimality.)

Which week i ? It does not matter: the fleet has no final week, so it looks the same from week 3 as from week 30, and every week gives the same average.

Remark 5.2. $Q^*(s, a)$ is different from $\mathbb{E}[G_i]$. The number $\mathbb{E}[G_i]$ averages over *all* executions of the policy, whatever state they happen to be in at week i . $Q^*(s, a)$ restricts the average to the executions that pass through s at week i and are made to play a there, acting optimally afterwards. And while $\mathbb{E}[G_i]$ is a single number, Q^* is a whole table of numbers, one per state-action pair: for the fleet, $4 \times 3 = 12$ entries.

The table is worth the trouble because the optimal policy can be read off it: at state s , the optimal action is simply the action a that maximizes $Q^*(s, a)$. In notation,

$$\pi^*(s) = \arg \max_a Q^*(s, a).$$

In words: in state s , compare the values $Q^*(s, a)$ of all available actions a and play an action with the largest value; $\arg \max$ denotes exactly that action, and π^* denotes the optimal policy.

It remains to compute the table. Suppose an optimal policy produced the trajectory $\dots, s_i, a_i, r_i, s_{i+1}, a_{i+1}, \dots$. Averaging both sides of the recurrence (4.3), $G_i = r_i + \gamma G_{i+1}$, over many such executions, it must be the case that

The Bellman equation

$$Q^*(s_i, a_i) = \mathbb{E}[r_i + \gamma Q^*(s_{i+1}, a_{i+1})]. \quad (5.1)$$

In words: the value of this week's state and action equals this week's reward plus γ times the value of next week's state and action, averaged over the randomness of the week. Replacing G_{i+1} by its average $Q^*(s_{i+1}, a_{i+1})$ is legitimate: averaging in two rounds (first among the executions that share next week's state and action, then across the week's outcomes) gives the same result, just as a company-wide average can be computed branch by branch.

We can therefore find Q^* by solving the Bellman equations, one equation per state-action pair, twelve for the fleet. Since the optimal policy at s_{i+1} picks the action with the largest Q^* -value, the right-hand side of (5.1) involves the same unknowns $Q^*(s, a)$ as the left-hand side, so this is a system of equations in the entries of the table. The visualization solves it iteratively; its Q^* -scorecard and Bellman-check scenes display the solution for the fleet at $\gamma = 0.9$.⁵ Two entries tell the story: $Q^*(\text{SHAKY}, \text{SERVICE}) = 126.4$, while $Q^*(\text{SHAKY}, \text{REPLACE}) = 149.9$. The optimal playbook replaces a van that still starts every morning.

Closing

These notes modeled sequential decision problems as Markov decision processes, let policies generate random trajectories, scored each trajectory by its cumulative reward, averaged that score to obtain something maximizable, and characterized the best policy through the Q -function and the Bellman equation. The fastest way to make all of this concrete is to play: open https://sml-fs26.github.io/classic_rl/repair-or-replace/, run the fleet for a quarter, drag the γ slider, and watch the Q^* scorecard fill in, scene by scene, in the same order as these notes.

⁵Scenes “ Q^* : the action scorecard” and “The Bellman equation,” interactive in the [visualization](#).